

Regression Model

byron

```
rm(list = ls()) #setup

# Loading packages

library(tidyverse)

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0    v readr      2.2.0
v forcats    1.0.1    v stringr    1.6.0
v ggplot2    4.0.2    v tibble     3.3.1
v lubridate  1.9.5    v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(performance) #model diagnostics

# Loading Data

human_cognitive_performance <- read.csv("D:/Behavioral Data Analysis/human_cognitive_performance_clean.csv",
                                         stringsAsFactors = F)

#Data inspection

str(human_cognitive_performance)

'data.frame':  80000 obs. of  14 variables:
 $ user_id      : chr  "U1" "U2" "U3" "U4" ...
 $ age          : int   57 39 26 32 50 36 48 35 29 57 ...
 $ gender       : chr   "Female" "Female" "Male" "Male" ...
 $ sleep_duration : num   6.5 7.6 8.2 7.8 9.7 7.4 6.7 6 5.1 7.9 ...
 $ stress_level  : int    3 9 6 9 2 10 2 1 6 9 ...
 $ diet_type     : chr   "Non-Vegetarian" "Non-Vegetarian" "Vegetarian" "Vegetarian" ...
 $ daily_screen_time : num   6.5 10.8 5.7 8.3 11.3 6.6 1.4 6.8 10.2 9.1 ...
 $ exercise_frequency: chr   "Medium" "High" "Low" "Low" ...
 $ caffeine_intake  : int   41 214 429 464 365 381 39 2 244 355 ...
 $ reaction_time    : num   583 368 445 286 238 ...
 $ memory_test_score : int    65 58 49 94 62 70 51 60 82 54 ...
 $ cognitive_score   : num   36.7 54.4 32.6 70.2 87.5 ...
 $ ai_predicted_score: num   39.8 57.7 29.5 74.6 91.8 ...
 $ age_group        : chr   "45-59" "30-44" "18-29" "30-44" ...

summary(human_cognitive_performance)

      user_id      age      gender      sleep_duration
Length:80000    Min.   :18.00  Length:80000    Min.    : 4.000
Class :character 1st Qu.:28.00  Class :character 1st Qu.: 5.500
Mode :character  Median :39.00  Mode  :character Median : 7.000
                  Mean   :38.53                Mean   : 7.005
                  3rd Qu.:49.00                3rd Qu.: 8.500
                  Max.   :59.00                Max.   :10.000

      stress_level  diet_type      daily_screen_time exercise_frequency
Min.   : 1.000    Length:80000    Min.   : 1.000    Length:80000
```

```

1st Qu.: 3.000   Class :character   1st Qu.: 3.800   Class :character
Median : 5.000   Mode  :character   Median : 6.500   Mode  :character
Mean   : 5.494                                     Mean   : 6.505
3rd Qu.: 8.000                                     3rd Qu.: 9.200
Max.   :10.000                                     Max.   :12.000
caffeine_intake reaction_time memory_test_score cognitive_score
Min.   : 0      Min.   :200.0   Min.   :40.0    Min.   : 0.00
1st Qu.:123     1st Qu.:300.1   1st Qu.:55.0    1st Qu.: 40.91
Median :249     Median :400.4   Median :70.0    Median : 58.36
Mean   :249     Mean   :400.0   Mean   :69.5    Mean   : 58.17
3rd Qu.:375     3rd Qu.:499.2   3rd Qu.:85.0    3rd Qu.: 75.83
Max.   :499     Max.   :600.0   Max.   :99.0    Max.   :100.00
ai_predicted_score age_group
Min.   : 0.00      Length:80000
1st Qu.: 40.85     Class :character
Median : 58.36     Mode  :character
Mean   : 58.12
3rd Qu.: 75.89
Max.   :100.00

```

```
names(human_cognitive_performance)
```

```

[1] "user_id"      "age"          "gender"
[4] "sleep_duration" "stress_level" "diet_type"
[7] "daily_screen_time" "exercise_frequency" "caffeine_intake"
[10] "reaction_time" "memory_test_score" "cognitive_score"
[13] "ai_predicted_score" "age_group"

```

```
#checking for missing variables
```

```

human_cognitive_performance %>%
  summarise(across(everything(), ~ sum(is.na(.))))

```

```

user_id age gender sleep_duration stress_level diet_type daily_screen_time
1      0  0      0              0              0              0
exercise_frequency caffeine_intake reaction_time memory_test_score
1              0              0              0
cognitive_score ai_predicted_score age_group
1              0              0

```

Variables of interest are age, gender, sleep duration, stress level, diet type, daily screen time, exercise frequency and caffeine intake.

```
#Setting up a new data frame with variables of interest
```

```

human_cognitive_performance1 <- human_cognitive_performance %>%
  select(age, gender, sleep_duration, stress_level, diet_type,
         daily_screen_time, exercise_frequency,
         caffeine_intake, cognitive_score)

```

```
str(human_cognitive_performance1)
```

```

'data.frame': 80000 obs. of 9 variables:
 $ age      : int  57 39 26 32 50 36 48 35 29 57 ...
 $ gender   : chr  "Female" "Female" "Male" "Male" ...
 $ sleep_duration : num  6.5 7.6 8.2 7.8 9.7 7.4 6.7 6 5.1 7.9 ...
 $ stress_level  : int  3 9 6 9 2 10 2 1 6 9 ...
 $ diet_type   : chr  "Non-Vegetarian" "Non-Vegetarian" "Vegetarian" "Vegetarian" ...
 $ daily_screen_time : num  6.5 10.8 5.7 8.3 11.3 6.6 1.4 6.8 10.2 9.1 ...
 $ exercise_frequency: chr  "Medium" "High" "Low" "Low" ...
 $ caffeine_intake  : int  41 214 429 464 365 381 39 2 244 355 ...
 $ cognitive_score  : num  36.7 54.4 32.6 70.2 87.5 ...

```

#From the data structure;Some of the variables are characters and would need to be recoded

```
#Recoding <chr> variables to factors
```

```
library(dplyr)

human_cognitive_performance1 <- human_cognitive_performance1 %>%
  dplyr::mutate(
    gender = dplyr::recode(gender,
                          "0" = "Male",
                          "1" = "Female")
  )

human_cognitive_performance1 <- human_cognitive_performance1 %>%
  dplyr::mutate(
    exercise_frequency = dplyr::recode(exercise_frequency,
                                       "1" = "Low",
                                       "2" = "Medium",
                                       "3" = "High")
  )

human_cognitive_performance1 <- human_cognitive_performance1 %>%
  dplyr::mutate(
    diet_type = dplyr::recode(diet_type,
                              "0" = "Non-Vegetarian",
                              "1" = "Vegetarian",
                              "2" = "Vegan")
  )

str(human_cognitive_performance1)
```

```
'data.frame': 80000 obs. of 9 variables:
 $ age      : int  57 39 26 32 50 36 48 35 29 57 ...
 $ gender   : chr  "Female" "Female" "Male" "Male" ...
 $ sleep_duration : num  6.5 7.6 8.2 7.8 9.7 7.4 6.7 6 5.1 7.9 ...
 $ stress_level  : int  3 9 6 9 2 10 2 1 6 9 ...
 $ diet_type   : chr  "Non-Vegetarian" "Non-Vegetarian" "Vegetarian" "Vegetarian" ...
 $ daily_screen_time : num  6.5 10.8 5.7 8.3 11.3 6.6 1.4 6.8 10.2 9.1 ...
 $ exercise_frequency: chr  "Medium" "High" "Low" "Low" ...
 $ caffeine_intake  : int  41 214 429 464 365 381 39 2 244 355 ...
 $ cognitive_score  : num  36.7 54.4 32.6 70.2 87.5 ...
```

Multiple Regression Model

```
mr_model <- lm(cognitive_score ~ age +
               gender +
               sleep_duration +
               stress_level +
               diet_type +
               daily_screen_time +
               exercise_frequency +
               caffeine_intake,
               data = human_cognitive_performance1)

summary(mr_model)
```

Call:

```
lm(formula = cognitive_score ~ age + gender + sleep_duration +
    stress_level + diet_type + daily_screen_time + exercise_frequency +
    caffeine_intake, data = human_cognitive_performance1)
```

Residuals:

Min	1Q	Median	3Q	Max
-48.13	-16.45	0.22	16.79	47.81

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    7.650e+01  4.864e-01 157.278  <2e-16 ***
age            -8.710e-03  6.057e-03  -1.438    0.150
genderMale     -4.420e-02  1.497e-01  -0.295    0.768
genderOther    -7.648e-02  3.775e-01  -0.203    0.839
sleep_duration  2.024e+00  4.226e-02  47.902  <2e-16 ***
stress_level   -1.853e+00  2.558e-02 -72.440  <2e-16 ***
diet_typeVegan -2.416e-01  2.520e-01  -0.959    0.338
diet_typeVegetarian -1.254e-01  1.636e-01  -0.767    0.443
daily_screen_time -1.450e+00  2.315e-02 -62.626  <2e-16 ***
exercise_frequencyLow -1.427e+01  2.004e-01 -71.235  <2e-16 ***
exercise_frequencyMedium -4.630e+00  2.003e-01 -23.117  <2e-16 ***
caffeine_intake -1.983e-02  5.071e-04 -39.092  <2e-16 ***

```

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 20.73 on 79988 degrees of freedom
Multiple R-squared:  0.1917,    Adjusted R-squared:  0.1916
F-statistic: 1725 on 11 and 79988 DF,  p-value: < 2.2e-16

```